x. It shall also be complaint with BS 476 Part 5, 6, 7 Class 0 and Part 8.

i) The flexible connectors shall allow freedom of movement without unnecessary slack and shall not be installed with tight twists.

j) When installed in a ductwork run, the metal edges of the flexible connection shall 40-50mm apart for optimum effectiveness in handling vibration and movement of the duct run.

k) Fabric part of flexible connectors shall not be painted.

l) Flexible connectors shall be protected with material such as baffle plate to act against the fluctuation air pressures and stresses due to the operation of the ventilation fans.

12. INSULATION FOR DUCT & COPPER PIPE

12.1 General Requirements

a) The section specifies the requirements for the vendor submittal, manufacturing, supply and installation of the insulation.

b) All insulation material delivered to site shall be new.

c) Thermal insulation shall be applied to all air conditioning distribution duct system, chilled water piping, equipment /vessel, valves and other fittings and refrigeration pipe and distribution system.

d) Air distribution systems conveying untreated fresh air need not be insulated except where they pass through air-conditioned space.

e) Fire rated insulation shall be provided for all fire rated duct system.

f) All Insulation materials shall be free from CFC, HFC and other Harmful substance which contribute to ozone depletion and global warming.

g) Aluminium cladding of 0.6 mm thick and the AL cladding shall be 2 coated of black colour paint with mat finish of supply and exhaust duct. The aluminium material shall follow the below specification.

Aluminium cladding	
Description	**Standard/Properties**
Standard	ASTM B 309/ IS 737
Type	Cold Role Al coil
Finish	Mill finish
Aluminium Alloy	AA 3003 / 31000
Temper	H14

h) Duct exposed to outside environment shall be provided with 15mm cement plaster finish on chicken-wire mesh with additional finish of 0.8mm polyisobutylene sheeting with solvent weld joints shall be applied to outdoor type insulation which is exposed to ambient wherever applicable. The whole shall be then provided with a completely weatherproof external enclosure to the insulation. Corrosion proof bands and clips shall hold the sheeting tight against the insulation such that ballooning will not occur.

i) 0.6 mm thick Stainless Steel (SS 316) chicken wire mesh of 25mm mesh size shall than be wrapped on the insulated duct to hold the insulation.

j) Adhesive used for the insulation shall be with low VOC content as IGBC guidelines.

k) All relevant certificates for all INSULATION shall be submitted to Engineer/Employer for NONO

12.2 Manufacturer's Eligibility criteria and Qualifications

a) Insulation material shall be manufactured in a factory / facility registered to ISO 9001 (Quality management system) and ISO 14001 (Environmental management system). The Valid Certification shall be submitted.

b) The manufacture shall show at least five years (5) continuous and current experience in the manufacturing and supply of Insulations of similar type proposed for the project as pre specification/ BOQ for period ending with did submission date.

c) The manufacturer (OEM) / contactor should have supplied, and installed Insulations for a minimum 50% of BOQ of similar type proposed for this project in any commercial / transportation / industrial domains and large infra structure projects. A job reference list to be submitted within the last five (5) years shall be submitted period ending bid submission date. The contractor/ OEM shall arrange visit to Employer/Engineer for the executed job as give in the reference list to obtain NONO.

12.3 Technical Specification of Product

12.3.1 Rockwool Insulation:

a) Rockwool insulation shall be provided wrapped with a factory applied Alu-glass vapour barrier jacketing for all fire rated ducts which is suitable for fire rated applications.
b) Testing of Rockwool insulation must comply with latest version of IS 8183, IS 3144, IS 3346.
c) Thickness of the insulation shall be of minimum 50mm.
d) Density of the insulation shall be greater than 105 Kg/m^3 as defined in BoQ.
e) Thermal conductivity (W/m°K) shall be limited to 0.034 maximum at 25°C mean temperature.
f) Working temperature range for Rockwool shall be between 0°C to 750°C.
g) Vapour barrier jacketing shall be of Aluglass Kraft paper, fire resistant adhesive and reinforced with fibre glass yarn. The Kraft paper shall be permanently treated to assure permanent fire and smoke safety, and to prevent corrosion of the foil.
h) The GSS strap and weld stick pin shall be provided as per relevant standards for fixing insulation.
i) The Alu-glass vapour barrier shall have the following physical properties:
   i. It shall comply with ASTM E 84. Flame spread index less than 25 and smoke developed (finished side) less than 50.
   ii. Water vapour Permeability (ASTM-E96): 1.14 mg/Ns
   iii. Beach puncture tear (ASTM-D-781-63T) : 1.15 kg m/m
   iv. Ultimate tensile Strength (TAPPI-404-TS-06) : 10.6 KN/m
   v. Bursting strength – 5.75 Kg/cm^2

12.3.2 Cross Linked Closed Cell Polyolefin (XLPE)

a) Thermal insulation material for Duct shall be Cross Linked Closed Cell Polyolefin (XLPE) which shall be UV resistant with factory made laminated of 140-micron thick fibreglass cloth grey/block in colour for mechanical protection.
b) Adhesive shall be provided by the same supplier of the insulation material.
c) Insulation thickness shall be of minimum 19 mm.
d) The Insulation material shall be fire rated for Class 0 as per BS 476 Part 6 : 1989 for fire propagation test and for Class 1 as per BS 476 Part 7, 1987 for surface spread of flame test.
e) Density of the Cross-Linked Closed Cell Polyolefin foam shall be 25-30 Kg/m3
f) Thermal conductivity as per BS EN 12667 / EN ISO8497 / ASTM C518 of the insulation material shall not exceed 0.036 W/mºK at an average temperature of 25oC. The thermal conductivity of insulation material shall not be affected by aging as per DIN 52616 standards (or) Equivalent standard.
g) Water vapour permeability shall be not less than 0.024 per inch (2.48 x 10-13 Kg/m.s.Pa i.e. μ>7000: Water vapour diffusion resistance) as per EN 12086 & EN13469.
h) The Material shall comply with ISO 5659 / BS 6853 / ABD 0031/ EN 45545-1 (R1 HL3) for smoke density and toxicity values.
i) The Insulation material shall be anti-microbial. Microbiological growth on the insulation surface shall be in accordance with ASTM G-21 and bacterial resistance to ASTM 2180 / ISO 22196.

j) Adhesive used shall be non-flammable and with low VOC content as per IGBC guidelines. Adhesive shall be provided by the same supplier of the insulation material
k) Tape shall use reinforced aluminium foil of minimum 9-micron thickness and minimum width 75 mm to provide vapour sealing of insulated ducts.
l) Sample and Mock-up shall be submitted to Engineer/Employer for notice of no objection.

12.3.3 Fiberglass Insulation:

a) Fiberglass insulation shall be provided wrapped with a factory applied Aluglass vapour barrier jacketing for all air conditioning supply ducts which his suitable for normal applications.
b) All testing of fiberglass insulation must comply with latest version of IS 8183, IS 3144,IS3346.
c) Thickness of the insulation shall be of minimum 50mm.
d) Density of the insulation shall be greater than 32 Kg/m^3 as defined in BoQ.
e) Thermal conductivity (W/m°K) shall be limited to 0.030 maximum at 10°C and 0.032 maximum at 25°C mean temperature.
f) Working temperature range for fiber shall be between 0°C to 230°C and foil face between 0°C to 100°C.
g) Vapour barrier jacketing shall be of Aluglass Kraft paper, fire resistant adhesive and reinforced with fibre glass yarn. The Kraft paper shall be permanently treated to assure permanent fire and smoke safety, and to prevent corrosion of the foil.
h) The GSS strap and weld stick pin shall be provided as per relevant standards for fixing insulation.
i) The aluglass vapour barrier shall have the following physical properties:
   i. It shall comply with ASTM E 84. Flame spread index less than 25 and smoke developed (finished side) less than 50.
   ii. Water vapour Permeability (ASTM-E96): 1.14 mg/Ns
   iii. Beach puncture tear (ASTM-D-781-63T) : 1.15 kg m/m
   iv. Ultimate tensile Strength (TAPPI-404-TS-06) : 10.6 KN/m
   v. Bursting strength – 5.75 Kg/cm^2
j) Non-combustible in accordance with BS 476 Part 4.
k) Insulation shall comply with Class O as per BS 476 Part 6 and 7.
l) Insulation shall comply with Class 1 as per BS 476 Part 7.
m) Sample shall be submitted to ER for notice of no objection.